MedAgent Clinical Decision Support

Case ID: case_767dbe7718 | **Status:** APPROVE | **Physician:** Dr. Alex Vance | **Date:** 2026-10-16 09:25 UTC

CRITICAL DISCLAIMER: Decision-support draft generated by AI agents. It is not a diagnosis and must be reviewed, corrected and approved by a licensed physician before any clinical use.

Patient Intake & Presentation

Patient: Age: 23, Sex: Male
Notes & History: .

Clinical Summary / Impression

Clinical evaluation indicates features consistent with Acute Kidney Injury, characterized pathophysiologically by abrupt decline in glomerular filtration rate leading to retention of nitrogenous wastes (azotemia), dysregulation of fluid balance, and impaired electrolyte and acid-base excretion. Key supporting clinical indicators include Renal Impairment, Hyperkalaemia, High Anion Gap Metabolic Acidosis, while secondary differential considerations include Severe Hyperkalemia, Metabolic Acidosis. Concurrently, chest radiography identifies Effusion (68.1%). Targeted diagnostic workup and guided serial clinical monitoring are recommended to establish definitive management.

[URGENT CLINICAL ALERT] RED FLAGS IDENTIFIED:
• Critical hyperkalemia with lethal arrhythmia risk. Urgent ECG and stabilization required.

Ranked Differential Diagnoses

#	Condition	Likelihood	Key Supporting Findings	Citations
1	**Acute Kidney Injury**	MODERATE	Renal Impairment, Hyperkalaemia, High Anion Gap Metabolic Acidosis	S7, S8
2	**Severe Hyperkalemia**	MODERATE	Hyperkalaemia, Renal Impairment, High Anion Gap Metabolic Acidosis	S5, S6
3	**Metabolic Acidosis**	MODERATE	High Anion Gap Metabolic Acidosis, Renal Impairment	S7
4	**Chronic Kidney Disease**	MODERATE	Renal Impairment, Hyperkalaemia	S_Chroni_1, S_Chroni_2
5	**Hyponatremia**	MODERATE	Hyponatraemia	S2

Radiographic Findings (Chest X-Ray)

Pathology / Finding	Probability	Location	Indicative Severity	Confidence
Effusion	**68.1%**	right lower zone	Moderate	Moderate

Laboratory Findings & Identified Patterns

Analyte / Test	Result	Unit	Reference Interval	Flag
Potassium	**6.5**	mEq/L	3.5 – 5	CRITICAL HIGH
eGFR	**10**	mL/min/1.73m2	60 – 130	CRITICAL LOW
Serum Creatinine	**4.8**	mg/dL	0.7 – 1.3	HIGH
BUN	**72**	mg/dL	7 – 20	HIGH
Sodium	**131**	mEq/L	135 – 145	LOW
Bicarbonate / CO2	**15**	mEq/L	22 – 29	LOW

Inferred Diagnoses (ClinicalLabNet Neural Classifier):
• **[SEVERE] Acute Kidney Injury / Renal Failure** [100.0% Neural Confidence] (Likelihood: HIGH): Neural Network Confidence: 100.0% (ClinicalLabNet deep multi-label classifier). Biomarker pattern is characteristic of acute kidney injury / renal failure. *Pathophysiology:* Abrupt decline in glomerular filtration rate leading to retention of urea, creatinine, and nitrogenous metabolic waste products with disruption of fluid, electrolyte, and acid-base homeostasis.
• **[CRITICAL] Severe Hyperkalemia** [96.2% Neural Confidence] (Likelihood: HIGH): Neural Network Confidence: 96.2% (ClinicalLabNet deep multi-label classifier). Biomarker pattern is characteristic of severe hyperkalemia. *Pathophysiology:* Dangerous elevation of extracellular potassium lowering myocardial resting membrane threshold, impairing cardiac repolarization, and predisposing to fatal ventricular arrhythmias or asystole.
• **[SEVERE] High Anion Gap Metabolic Acidosis** [59.7% Neural Confidence] (Likelihood: LOW): Neural Network Confidence: 59.7% (ClinicalLabNet deep multi-label classifier). Biomarker pattern is characteristic of high anion gap metabolic acidosis. *Pathophysiology:* Pathological accumulation of unmeasured fixed organic acids (lactic acid, ketoacids, uremic toxins) consuming serum bicarbonate buffers.

Synthesized Clinical Patterns:
• **[URGENT] Renal Impairment:** Elevated nitrogenous wastes and/or reduced filtration rate indicating acute or chronic renal dysfunction.
• **[URGENT] Hyperkalaemia:** Elevated serum potassium conferring risk of fatal cardiac arrhythmias.
• **[SOON] Hyponatraemia:** Subnormal serum sodium requiring cautious osmolality evaluation and rate-controlled correction.
• **[SOON] Metabolic Acidosis:** Decreased serum bicarbonate / total CO2 indicating systemic acid accumulation or bicarbonate loss.

Suggested Diagnostic Workup

• Renal ultrasound to exclude hydronephrosis
• Urine electrolytes and fractional excretion of sodium
• Discontinue nephrotoxic agents
• Immediate 12-lead ECG for peaked T waves/conduction blocks
• Calculate serum anion gap [Na - (Cl + HCO3)]
• Spot urine albumin-to-creatinine ratio
• Serum and urine osmolality

Physician Attestation: I, Dr. Alex Vance, hereby confirm that I have reviewed, verified, and adjusted this decision-support synthesis. Decision: **APPROVE**.
Reviewer Notes: No specific notes recorded.
Signature: _____ **Date:** 2026-10-16 09:25 UTC